

APPENDIX III: Calculation of the Rate of Surplus-Value
in the U.S. - 1952-1980.

A.III.1.

The purpose of this Appendix is to outline the method of the calculation of the rate of surplus-value in the U.S. for the period 1952-1980. The empirical basis of this calculation is the National Income and Product Accounts (NIPA) which is itself compiled from the perspective of Keynesian economics.

It is known that Marxian categories such as constant capital (c), variable capital (v) and surplus-value (s) belong to a high level of abstraction - not present in Keynesian categories. Therefore, in order to measure the rate of surplus-value (s/v), the theoretical transformation of categories from Keynesian to Marxian ones is required. So far there have been two main attitudes in approaching the measurement of the rate of surplus-value.¹ The first approach involves an explicit or implicit assumption that there exists a direct correspondence between Keynesian and Marxian categories,² which thus directly applies data based upon the Keynesian framework in quantifying Marxian concepts, e.g. the rate of surplus-value (s/v). Challenging the preceding approach, some Marxists feel that it is not possible, in principle, to derive any empirical content from Marxian value categories. However, there is another and rarely used theoretical approach upon which

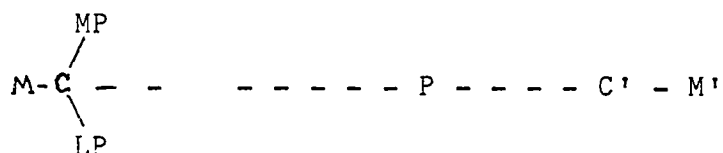
Gillman (1957), Mage (1967) and Shaikh (1978a) have based their works. The importance of the theoretical understanding common to these studies is the emphasis they have placed upon the distinction between value and forms of value. These authors, first of all, discuss the correspondence between statistical data which is based on the Keynesian framework and the money forms of Marxian value categories. Only after the necessary theoretical discussion of statistical data and their transformation into money forms of value, do they attempt to measure the Marxian categories and ratios.³

In the second part of the Appendix, the fundamental concepts such as constant capital, variable capital, surplus-value and the rate of surplus-value are theoretically discussed. In the third part, the empirical procedures of the calculation of variable capital and that of surplus-value are presented. The fourth part presents these procedures in detail and step-by-step fashion by utilizing 1964 data while the fifth section mainly consists of the tables of raw data and other related measurements.

A.III.2.

The intention of capitalist production is, first and foremost, to make money by merely advancing money. Marxian economics and its categories differ from mainstream economics in analyzing this apparent fact.

Within the main-stream tradition, it is claimed that all "factors" involved in production obtain their share of total value according to their marginal productivity. Capital receives profits while labor receives wages, and the end result of the distribution of produced wealth is considered fair. Within the Marxist tradition, on the other hand, the fact of "making more money by advancing money" is analyzed in the context of the circuit of money capital which is represented by the following scheme:



In this circuit, (M) demotes the necessary amount of advanced money and the beginning of capitalist production which buys necessary commodities (C) for production, i.e. labor-power (LP) and the means of production (MP). (P) represents the production process through which inputs (C) are transformed into outputs (C') which themselves embody more labor time than inputs. (M) and (M') are therefore money forms of the mass of embodied labor time in (C) and (C') respectively; with the assumption that the proportionality between values and prices (M') is larger than (M). The difference between the initial money form of capital (M) and the final money form of capital (M'), i.e., $M' - M = \Delta M$, is exactly what Marx calls the money-form of surplus-value.

This is Marx's solution to the mystery that lies behind "making money by merely advancing money": "Industrial capital encompassing every branch of production that is pursued on the capitalist basis is the only mode of existence of capital in which not only the appropriation of surplus-value or surplus product, but also its creation, is a function of capital. It thus requires production to be capitalist in character; its existence includes that of the class antagonism between capitalists and wage-labourers..." (Marx, 1893, pp. 135-6)

Within the general circuit of capital $M - C \dots P \dots C' - M'$, means of production participate in the production process in the form of some commodities in which a certain amount of labor is embodied, and they can transfer only that amount of labor into newly produced commodities.⁴ This part of commodity capital (C) is termed constant capital (C) since it does not contribute to the change in labor time, $C = C' - C$, but merely transfers its labor content. The other part of (C) is the commoditized labor-power which "gains" this feature, i.e. being a commodity, under the capitalist mode of production so that it can be bought and sold like other commodities. Marx defines labor power as "the aggregate of those mental and physical capabilities existing in the human being, which he exercises whenever he produces a use-value of any description."⁵ Labor power, a unique

commodity, becomes the sole creator of surplus-value in some sense due to the logic of the capitalist mode of production, through the division of working day into necessary and surplus parts and freeing workers from the means of production. Nevertheless, its uniqueness does not imply that the value of labor-power is determined in different ways. On the contrary, its value is determined, as in the case of other commodities, by the labor time socially necessary for its reproduction. Marx calls the part of the money capital (M), which is paid to the owner of the labor-power, the variable capital (v). It is termed variable capital, because this unique commodity is employed for a certain time period, which is longer than the necessary time than is required for reproduction of labor-power, and produces the corresponding value increment M, during the surplus-labor time. At the end of the production process, as a result of the expenditure of surplus-labor time, the value of inputs, (c+v) increases by the amount of embodied surplus-labor time (s) which yields the value of output, (c+v+s).

In making use of necessary labor time, (v), and surplus-labor time, (s), it is possible to define the rate of surplus-value by constructing the ratio of the surplus-value to variable capital, (s/v). The rate of surplus-value is also a measure of degree of exploitation of labor since it expresses the ratio of unpaid to

paid labor.⁶

Changes in the rate of surplus-value reflect and affect class struggle. All attempts by capital to increase and by labor to reduce this rate sharpen the class contradiction. There are two fundamental ways of increasing the rate of surplus-value. The first method is based on the lengthening of the working-day; the production of absolute surplus-value. For example, let us assume that the necessary part of the working-day, say four hours, is constant while the working-day's length changed from eight hours to ten hours. Then, the rate of surplus-value increases from 100 per cent, $(s/v)=(4/4)$, to 150 per cent, $(s/v)'=(6/4)$. The second method is known as the production of relative surplus-value, which means to shorten the necessary labor time while keeping the working-day constant. For example, the reduction of necessary labor time from four hours to three hours in an eight hour working-day corresponds to the increase in the rate of surplus-value from 100 per cent, $(s/v)=4/4$, to 166 per cent, $(s/v)'=(5/3)$. Marx especially emphasizes the relative surplus-value production and its forms in increasing the rate of surplus-value, as capitalism develops.

A.III.3 Measurement of the Money form of Variable Capital and of Surplus-Value

To measure the rate of surplus-value obviously necessitates knowing the magnitude of variable capital

and surplus-value. In order to calculate these categories, first the "industries" must be classified according to the theoretical distinction between productive and unproductive labor/activities as discussed in Chapter III. In our classification, the productive sector consists of agriculture, forestry and fisheries, mining, construction, manufacturing, transportation and public utilities, and government enterprises. Additionally, a certain group of activities in the service industry is also considered as a part of the productive sector; namely, hotels and other lodging places, personal services, auto repair services and garages, miscellaneous repair services, motion pictures, and amusement and recreation services.

The unproductive sector, on the other hand, consist of wholesale and retail trade, finance, insurance and real estate, and government industries, and certain branches of service industry, namely business services, health services, legal services, educational services, social services and membership organizations, miscellaneous professional services, and private households.

In our empirical procedure, the classification of industries as productive and unproductive is especially significant regarding the calculation of variable capital since the magnitude of surplus-value is calculated as a residual of total value-added.

Variable Capital

In calculating the magnitude of variable capital, our main data sources are Employment and Training Report of the President (ETRP) and NIPA. The method of calculation is itself a four-step procedure. The first step involves determining the ratio of productive workers to the total labor force in that particular industry. The second step is the calculation of productive workers' share in employers' contributions for social insurance by using the ratio developed in the previous step as proxy. The Third step involves calculating annual salaries of productive workers on the basis of their average weekly wages with a 52-week employment assumption. It must be emphasized that the latter assumption of a 52-week employment duration most probably would lead to inflated figures for variable capital, hence conservative magnitudes for the rates of surplus-value. As a last step, the annual salaries of productive workers are added to their share in employers' contributions for social insurance to obtain total wages of productive workers, namely total variable capital in that industry. This outlined procedure, with some slight modifications which are described in Table I, is applied to manufacturing, mining, construction, transportation, utilities and communication, above-mentioned productive branches of service industry,

government enterprises and agriculture. To sum up, the procedure can be presented in the following manner:

Step I: Constructing the ratio of the number of productive workers to the total labor force in that industry. (A)

Step II: To determine the share of productive workers (C), employer contributions for social insurance (B) are multiplied by the ratio (A), which is considered as proxy. $(A) \times (B) = (C)$

Step III: Using average weekly earnings and the assumption of 52-week employment, the total annual salary of productive workers is determined. (D)

Step IV: Total variable capital in the industry is calculated by adding annual salaries of productive workers (D) to the share of productive workers in employers' contributions for social insurance (C). $(C) + (D)$

Surplus-value

As previously noted, the magnitude of surplus-value is calculated as a residual of total value-added. The procedure itself is relatively simple compared to other suggested methods of calculating the surplus-value and has only four steps. First, in order to obtain unadjusted total value-added figures, capital consumption allowances with capital consumption adjustments are subtracted from total gross domestic product. The next two steps involve two necessary

adjustments due to the nature of the NIPA framework. The first adjustment is related to the fictitious elements which do not represent any value production in the economy; namely imputations regarding rental, government, financial and household sectors. The second adjustment is required because of NIPA's peculiar treatment of surpluses of government-owned liquor stores as a government expenditure item. The last phase of our calculation of the magnitude of surplus-value involves subtracting variable capital from adjusted value-added which yields surplus-value itself.

The above procedure can be outlined as follows:

Step I: Obtaining total unadjusted value-added, (C) requires subtracting capital consumption allowances (B) from total gross domestic product (A). $(A)-(B)=(C)$

Step II: The first set of fictitious elements related to various rent imputations with capital consumption allowance adjustments yields total imputed rent (D). Then, the second set of fictitious elements related to financial (E), government (F), and household (G) sectors is added to (D) yielding total exclusions (H).

$$(D)+(E)+(F)+(G)=(H)$$

Step III: The surplus of government-owned liquor stores (I) is added to total unadjusted value-added (C) yielding adjusted value-added (J).

$$(C)+(I)-(H)=(J)$$

Step IV: Subtracting previously calculated variable-